INCH-POUND
MIL-DTL-23053/6D
17 June 1996
SUPERSEDING
MIL-I-23053/6C
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYOLEFIN, SEMI-RIGID, CROSSLINKED

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053

REQUIREMENTS:

Polymer type: The base polymer used in formulating this sleeving shall be a polyolefin.

Continuous operating temperature range: -55°C (-67°F) to +135°C (+275°F).

Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2a):

Class 1 - Flame resistant
Class 2 - Flammable (clear only)

Color: Class 2 shall be furnished in clear. Class 1 shall be furnished in colors only. Unless otherwise specified by the acquisition activity (see 6.2b), each size of Class 1 sleeving shall be furnished in the color designated in Table I. All colors shall conform to Class 1 of MIL-STD-104.

Longitudinal change: ± 5 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

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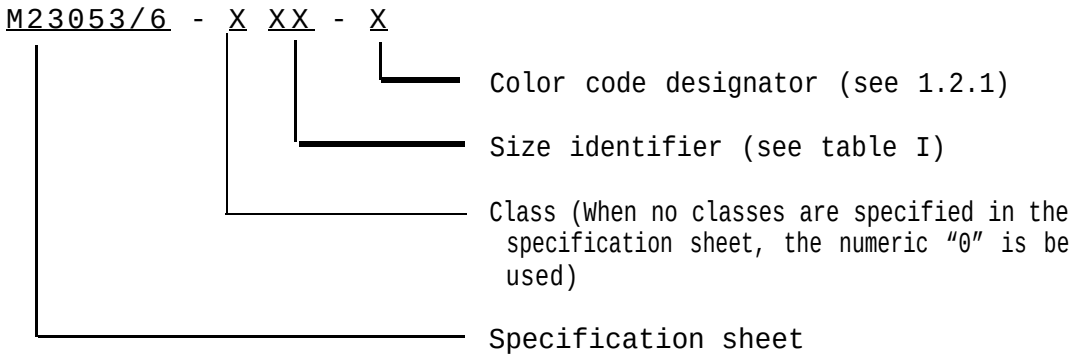

Example: Class 2, 0.125 inch (3.18 mm) as supplied ID sleeving shall be identified as M23053/6-204-C.

TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall Thickness 2/
Class 1			
M23053/6-101-4	.046(1.17)	.023(.58)	.020 ± .003(.508 ± .076)
M23053/6-102-9	.063(1.60)	.031(.79)	.020 ± .003(.508 ± .076)
M23053/6-103-6	.093(2.36)	.046(1.18)	.020 ± .003(.508 ± .076)
M23053/6-104-2	.125(3.18)	.062(1.59)	.020 ± .003(.508 ± .076)
M23053/6-105-5	.187(4.75)	.093(2.36)	.025 ± .003(.635 ± .076)
M23053/6-106-8	.250(6.35)	.125(3.18)	.025 ± .003(.635 ± .076)
M23053/6-107-1	.375(9.53)	.187(4.75)	.030 ± .005(.762 ± .127)
M23053/6-108-9	.500(12.7)	.250(6.35)	.030 ± .005(.762 ± .127)
Class 2			
M23053/6-201-C	.046(1.17)	.023(.58)	.020 ± .003(.508 ± .076)
M23053/6-202-C	.063(1.60)	.031(.79)	.020 ± .003(.508 ± .076)
M23053/6-203-C	.093(2.36)	.046(1.17)	.020 ± .003(.508 ± .076)
M23053/6-204-C	.125(3.18)	.062(1.58)	.020 ± .003(.508 ± .076)
M23053/6-205-C	.187(4.75)	.093(2.36)	.025 ± .003(.635 ± .076)
M23053/6-206-C	.250(6.35)	.125(3.18)	.025 ± .003(.635 ± .076)
M23053/6-207-C	.375(9.53)	.187(4.75)	.030 ± .005(.762 ± .127)
M23053/6-208-C	.500(12.7)	.250(6.35)	.030 ± .005(.762 ± .127)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data

2/ Wall thickness values are less when shrinkage is restricted.

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3/ The color codes cited above for Class 1 sleevings are preferred acquisition colors. Class 1 sleevings are also available in other colors (see 1.2.1)

Unrestricted shrinkage: Test method 4.6.5, 200° ± 2°C (392° ± 4±F) for 3 minutes.

TABLE II. Physical properties. 1`/

Characteristic	Requirement	Test procedure and condition
As supplied:		
ID, minimum	Table I	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 250° ± 3°C (482° ± 6°F)
Secant modulus, psi (MPa), minimum	25,000 (172.4)	4.6.12.1, 2 percent strain, ASTM D882
Color stability	Pass	4.6.15 175° ± 2°C (347° ± 4°F), 24 hours
Concentricity	70% minimum	4.6.3.3
Clarity stability, Class 2	Pass	4.6.16, 175° ± 2°C (347° ± 4°F), 24 hours
Restricted shrinkage	No cracks	4.6.6.1.1 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	2,000(13.8)	4.6.13 ASTM D638, 2 inches/minute

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TABLE II. Physical Properties. 1/ - Continued

Characteristic	Requirement	Test procedure and condition
Ultimate elongation, percent, minimum	200	4.6.13 ASTM D638, 2 inches/minute
Dielectric strength volts/mil (Kv/mm), minimum	500 (19.7)	4.6.2 ASTM D2671
Volume resistivity, Ohm-cm, minimum	10^{14}	4.6.2. ASTM D876
Dielectric constant, maximum 3/	Class 1 - 3.2; Class 2 - 2.7	4.6.2 ASTM D150
Specific gravity, maximum	Class 1 - 1.35; Class 2 - 1.00	4.6.2 ASTM D792
Water absorption, percent, maximum	Class 1 - 0.5; Class 2 - 0.2	4.6.2 ASTM D570, 24 hrs at 23°C
Corrosion	No corrosion	4.6.10.1 and 4.5.10.2, 150° ± 2°C (302° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 1°C (-67° ± 2°F)
Heat resistance, properties after:		4.6.9 175° ± 2°C (347° ± 4°F) for 168 hours
Ultimate elongation, percent, minimum	100	4.6.11
Fluid resistance, properties after:		
Tensile strength, psi (MPa), minimum	1,600(11.1)	
Dielectric strength, volts/roil (Kv/mm), minimum	400 (15.8)	

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Flammability	Class 1 only - self-extinguishing within one minute and no more than 25 percent of in-dicator flag burnedor charred. No dripping or flowing of large sizes; Class 2 - N/A	4.6.14 Procedure B ASTM D2671
Fungus resistance 2/	No Growth	4.6.2 ASTM G21

1/ Unless otherwise specified, the requirements, test procedures and conditions are for all classes.

2/ MIL-I-23053/6 materials do not normally support fungus growth. Performance of this requirement is at the option of the acquiring activity (see 6.2).

3/ Dielectric constant is a requirement only when specified in acquisition document (see 6.2).

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 5 years. Conformance to 3.5. See 3.5.2 for storage life extension.

NOTE:

a. Flammability - Class 2 heat shrinkable sleeving described in this specification sheet have not been flame retarded and will burn readily. These sleevings (Class 2) shall not be considered for acquisition when flame resistance is an essential requirement.

Intended use: Heat shrinkable semi-rigid polyolefin sleeving is used in wire identification by marking or strain relief applications.

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Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA, ME, MI
Navy - EC, MC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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